

Study Guide for Test #2

Note: This study guide is meant to help you pinpoint specific topics that will be on the upcoming test. All information covered in lecture and assigned readings is fair game for the test regardless of whether it is mentioned specifically on this study guide.

Neurotransmitters and their Receptors

Compare and contrast the two main types of receptors

What are the two major classes of NTs?

How do these classes differ in terms of synthesis, storage, release, and inactivation?

For the major NTs discussed in class (Glu, GABA, Ach, DA, NE/Epi, 5-HT), describe the location of where it is synthesized, the precursors and enzymes important for synthesis, enzymes important for inactivation and possible transporter proteins, and receptors.

Also, describe what kind of behavior/effect increasing levels of the substance is associated with (e.g. GABA, drowsy)

Molecular Signaling Within Cells

Compare and contrast the different types of metabotropic receptors

List the 7 broad steps of a typical signaling cascade.

How do these steps lead to signal amplification?

What are the two classes of G-protein coupled receptors?

Describe the activation/inactivation cycle of each

List some common effectors and what they do.

List some common 2nd messengers and what they do.

Compare and contrast kinases and phosphatases.

What are the 3 transcriptional activator proteins discussed in class?

How do these become active?

What is the result of it becoming active?

Describe NGF's signaling pathways

Describe Tyrosine Hydroxylase's signaling pathways

Synaptic Plasticity

What is synaptic plasticity?

What are considered short-term changes? Long-term changes?

What is synaptic facilitation?

How does it work?

How long does it last?

What is synaptic depression?

How does it work?

How long does it last?

Why is Aplysia famous in memory research?

Describe the neural circuits important for memory.

What is habituation?

What causes habituation to occur?

What happens on a cellular level as a result?

How are the AP of the presynaptic cell and the EPSP of the postsynaptic cell affected?

What is sensitization?

What causes sensitization to occur?

What happens on a cellular level as a result? Short-term? Long-term?

How are the AP of the presynaptic cell and the EPSP of the postsynaptic cell affected?

Compare and contrast long-term potentiation and long-term depression

What causes them to occur?

How are PSPs altered?

What changes happen at the cellular level?

How is the morphology of the postsynaptic cell changed?

What is synaptic pruning?

What causes synaptic pruning to occur?

Why might it be the case that synaptic pruning is a normal part of development?

What happens to the brain and behavior in cases where synaptic pruning is decreased?

What are the core symptoms of Autism spectrum disorder?

How might brain abnormalities be linked to some of the symptoms?

Touch

How do receptor potentials differ from postsynaptic potentials and action potentials?

What does it mean for a sensory receptor to be a *mechanoreceptor*?

List the different receptor types for our sense of touch.

Describe the different touch receptors and the type of information they send to the brain.

What kind of stimulus activates the receptor?

In which layer of the skin and which type of skin is the receptor typically found?

What is a receptive field? How do receptive fields differ for different types of touch receptors?

What is innervation density? How does innervation density differ for different types of receptors (especially in parts of the body associated with fine touch)?

Is the receptor slow adapting or rapid adapting?

What does this mean?

Through what part of the sensory neuron does a receptor potential travel and potentially trigger an action potential?

Where can you find the cell body for a sensory neuron?

What types of fibers carry touch information to the spinal cord?

How are proprioceptors similar, but different in comparison to touch receptors?

Describe the different proprioceptive receptors and the type of information they send to the brain.

What types of fibers carry proprioceptive information to the spinal cord?

What are the differences between afferent fibers and efferent fibers in the spinal cord?

What are the differences between the dorsal horn and the ventral horn in the spinal cord?

How are inputs and outputs organized in each?

What is a dermatome?

What are the names of the different touch and proprioceptive pathways through the spinal cord?

Be able to draw each pathway.

What are the major synapse sites along the path (think primary, secondary, tertiary afferents)

Where do the fibers cross the midline of the body?

How do the paths differ if information is coming from the face versus the rest of the

body?

Describe the different parts of the brain that are important for touch and proprioception.

What type of input does each area receive?

Does the area interact with any other parts of the brain?

Describe how somatosensation is impaired in conditions such as multiple sclerosis, autism spectrum disorder, sensory processing disorder, and Parkinson's disease.

Pain

Define pain

How does chronic pain differ from acute pain

What can happen to the nervous system over time with chronic pain?

What is the primary receptor type for pain/temperature?

What kind of stimulus activates the receptor?

What types of fibers send pain information to the brain?

Describe how different fiber types are associated with speed of transfer, quality of pain, and behavioral response.

What is the name of the pathway that sends pain information to the brain?

Be able to draw the pathway.

What are the major synapse sites along the path (think primary, secondary, tertiary afferents)

Where do the fibers cross the midline of the body?

How do the paths differ if information is coming from the face versus the rest of the body?

What is the pain matrix?

Describe the two different pathways of the pain matrix. How do they differ anatomically and in terms of perception of pain?

Be able to draw the descending pathway that can modulate pain information.

Which parts of the brain are at the top of the hierarchy?

How do the parts of the brain involved with top-down control of pain relate to the pain matrix?

What role does the periaqueductal gray area play?

What types of chemicals are these areas sensitive to?

Describe the two mechanisms of gate control theory.

How does the placebo effect work to relieve pain?

Compare and contrast peripheral sensitization and central sensitization.